# Relational model and 4 types of constrains

**<u>Introduction to Relational model:</u>**

- The relational model is a fundamental concept in the database design.
- It represents the data as tables with well-defined relationships
- Each tables has rows known as tuples and columns known as attributes
- The relational model provides a simple, yet powerfull ways to manage data

**<u>Constrains and types of constrains:</u>**

**<u>CONSTRAINS</u>**: The constrains ensures the data consistency and integrity.

**<u>Types:</u>**

- Domain constrains: it defines the allowed values for the column or attribute
- Entity constrains:it ensures each row is unique
- Referential constrains:
    1. it defines the relationship between the two tables.
    2. Foreign key references a primary key in another table.
- Key constrains:
    1. It defines uniqueness and relations
    2. Primary key must be unique not null
    3. Unique key must be unique not null

**Constraints help maintain data integrity , reduce the errors and improve the data quality**

**Summary:**

The relationship model is a powerfull way to manage the data Where the constrains ensures the data consistency and integrity, understanding the those concepts is crucial for for effective database design and management .